

DATABASE DESIGN ANALYSIS

ASSIGNMENT 3

DATABASE MANAGEMENT SYSTEM

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INTRODUCTION

Database design analysis is an important process in developing an efficient and reliable database system. It helps identify data redundancy, insertion anomalies, deletion anomalies, and update anomalies. It also involves identifying functional dependencies and improving the structure of tables. In this assignment, the Online Shopping Management System is analysed to maintain accurate customer, product, order, payment, and delivery information.

DATA REDUNDANCY AND POTENTIAL ANOMALIES

DATA REDUNDANCY

Data redundancy occurs when the same information is stored repeatedly. It increases storage requirements and can cause inconsistent records.

In an online shopping system, customer addresses, supplier contact details, and product category names could be repeated in many order records.

For example, if a supplier's phone number is copied into every product row, changing the phone number requires updating several rows.

The proposed design reduces redundancy by storing customers, categories, products, suppliers, orders, order items, payments, and deliveries in separate tables.

INSERTION ANOMALY

An insertion anomaly occurs when information cannot be added without inserting unrelated information. If product and order details were stored in one table, a new product might not be registered until a customer ordered it. Improvement: store product information independently in the PRODUCT table and connect it to CATEGORY and SUPPLIER through foreign keys.

DELETION ANOMALY

A deletion anomaly occurs when deleting one record unintentionally removes other useful information. If supplier details were stored only alongside products, deleting the supplier's last product could also remove the supplier information. Improvement: maintain a separate SUPPLIER table.

UPDATE ANOMALY

An update anomaly occurs when the same fact is repeated and every copy must be changed. If a category name is repeated in many product records, renaming the category may leave inconsistent values. Improvement: store category details once in CATEGORY and refer to it using Category_ID.

FUNCTIONAL DEPENDENCIES BETWEEN ATTRIBUTES

A functional dependency $X \rightarrow Y$ means that attribute X uniquely determines attribute Y. The primary key in each table determines the other attributes in that table.

Table	Functional Dependencies
CUSTOMER	customer_id → customer_name, email, phone, address
CATEGORY	category_id → category_name, description
PRODUCT	product_id → product_name, price, stock_quantity, category_id, supplier_id
SUPPLIER	supplier_id → supplier_name, email, phone
ORDERS	order_id → customer_id, order_date, total_amount, order_status
ORDER_ITEM	order_item_id → order_id, product_id, quantity, unit_price
PAYMENT	payment_id → order_id, payment_date, payment_method, payment_status
DELIVERY	delivery_id → order_id, delivery_date, delivery_address, delivery_status

HOW POOR DATABASE DESIGN CAN LEAD TO ANOMALIES

Poor design causes duplicate data, inconsistent records, and difficulties inserting, updating, or deleting information. Repeating customer details in every order may lead to different phone numbers for the same customer. Combining products and suppliers may also make supplier records difficult to retain when a product is removed. These problems reduce accuracy and increase maintenance effort.

PROPOSED IMPROVEMENTS TO THE DATABASE DESIGN

1. Separate entities into CUSTOMER, CATEGORY, PRODUCT, SUPPLIER, ORDERS, ORDER_ITEM, PAYMENT, and DELIVERY tables.
2. Use primary keys to identify each record and foreign keys to connect related tables.
3. Apply normalization to reduce repeated facts and prevent insertion, deletion, and update anomalies.
4. Use UNIQUE constraints for customer and supplier email addresses where required.
5. Use CHECK constraints to prevent negative prices, stock quantities, quantities ordered, and order totals.

CONCLUSION

The Online Shopping Management System uses a structured relational design to manage shopping operations. Separating related information into appropriate tables reduces redundancy, prevents anomalies, and improves data consistency. Primary keys, foreign keys, and integrity constraints provide a reliable foundation for database operations.